SIEMENS

SIMATIC

ET 200SP HA
Analog Input Module AI 16xI 2-wire HART HA (6DL1134-6TH00-0PH1)

Manual

08/2019
A5E39408968-AC

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Security information 1

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Preface 2

Validity of the documentation

This manual describes the AI 16xI 2-wire HART HA I/O module with article number 6DL1134-6TH00-0PH1.

It supplements the "ET 200SP HA Distributed I/O System" system manual.

Functions that generally relate to the system are described in this manual.

The information in this manual and in the system and function manuals enables you to commission the ET 200SP HA.

Appendices

The appendices provide information that is relevant for using the ET 200SP HA outside the PCS 7 environment. If you do not plan to use the ET 200SP HA outside the PCS 7 environment, you will not need the information in the appendices.

Conventions

Please also observe notes marked as follows:

Note

A note contains important information on the product described in the documentation, on the handling of the product and on the section of the documentation to which particular attention should be paid.

Changes compared with the previous version

The following is an overview of the most important changes in the documentation compared to the previous version:

As of firmware V1.1.0:

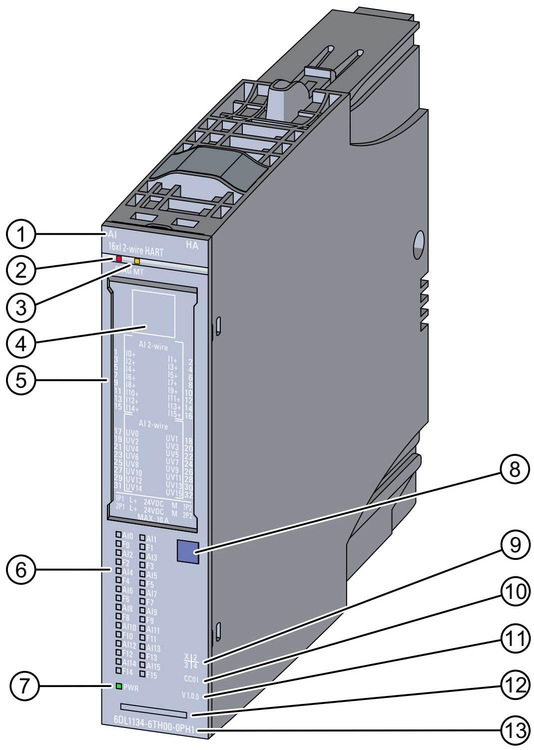
- Failure monitoring according to NE43 in the measuring range 4...20 mA replaces the configurable wire break limit.
  If the I/O module is configured as a firmware V1.0 variant with firmware V1.1 or higher, it supports the configurable wire break limit of V1.0 instead of failure monitoring.
  You can find information on failure monitoring in the section "Explanation of the module/ channel parameters (Page 18)".

Product overview

3

3.1 Properties of the AI 16xI 2-wire HART HA I/O module

View of the module



① Module type and designation
② LED for diagnostics
③ LED for maintenance
④ 2D matrix code
⑤ Wiring diagram
⑥ LEDs for channel status/channel fault
⑦ LED for supply voltage
⑧ Module type color coding
⑨ Product version
⑩ Function status and firmware version
⑪ Color code for selection of the color-coded labels
⑫ Serial number
⑬ Article number

Figure 3-1 View of AI 16xI 2-wire HART HA

Properties

The I/O module has the following technical properties:

- 16 analog inputs
- 16 outputs as encoder supply for 2-wire transmitters

- The analog inputs have the following channel-specific parameterizable properties:
  - Current measurement type for 2-wire HART transmitter
  - Measuring range 0 to 20 mA, 0 to 10 mA, 4 to 20 mA and 4 to 20 mA with HART
  - Resolution dependent on measuring range and interference frequency suppression, minimum 15 bits including sign, maximum 16 bits including sign
  - Smoothing
  - Interference frequency suppression 10 Hz, 50 Hz and 60 Hz
- Channel-specific configurable diagnostics
- Configurable failure monitoring (according to NE43)
- Module-specific configurable diagnostics for missing supply voltage L+
- Hardware interrupt at limit violation, per channel (two high and two low limits each)

The module supports the following functions:

- Firmware update
- I&M identification data
- Parameter reassignment in RUN
- Value status QI
- IO redundancy
- HART communication (Rev. 5 to Rev. 7)
- Up to 8 HART variables directly in the input address area
- A multiHART area in the input/output area

You can configure the I/O module with HW Config and integrate it into your system.

Accessories

The following accessories must be ordered separately:

- Labeling strips
- Color-coded labels
- Reference identification label
- Shield connector

Wiring 4

4.1 Terminal blocks for the I/O module

You can operate the I/O module with the following terminal blocks:

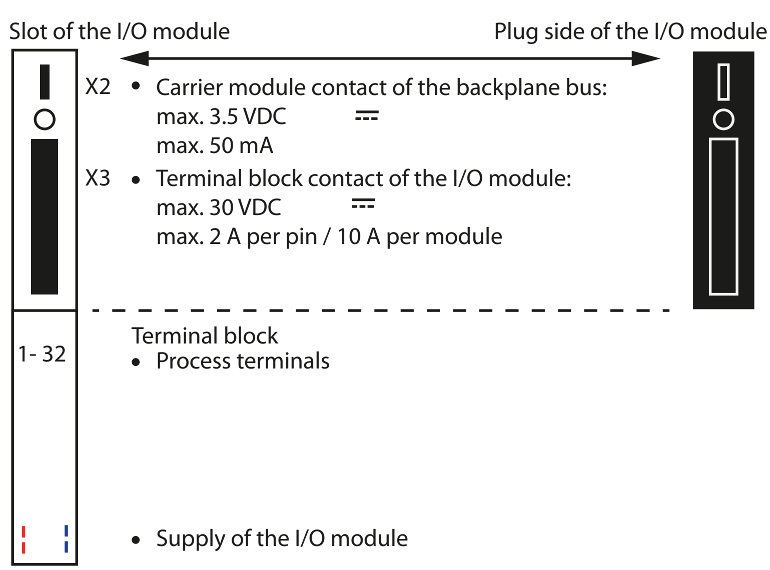
- 6DL1193-6TP00-0DH1, light
- 6DL1193-6TP00-0BH1, dark
- 6DL1193-6TP00-0DM1, light, for redundant configuration
- 6DL1193-6TP00-0BM1, dark, for redundant configuration
- 6DL1193-6TP00-0DN0, light, with additional potential distributors for ground connection
- 6DL1193-6TP00-0BN0, dark, with additional potential distributors for ground connection

Terminal blocks are not included in the scope of delivery of the I/O module and must be ordered separately.

Note

You can find additional information on the configuration in the system manual.

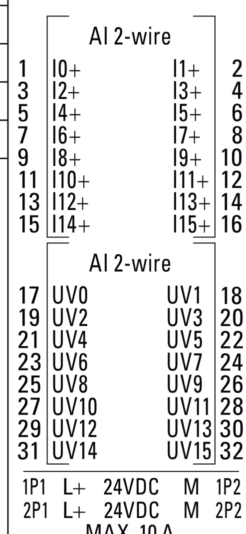
Connections on the slot and I/O module



4.2 Pin assignment of the AI 16xI 2-wire HART HA I/O module

General pin assignment

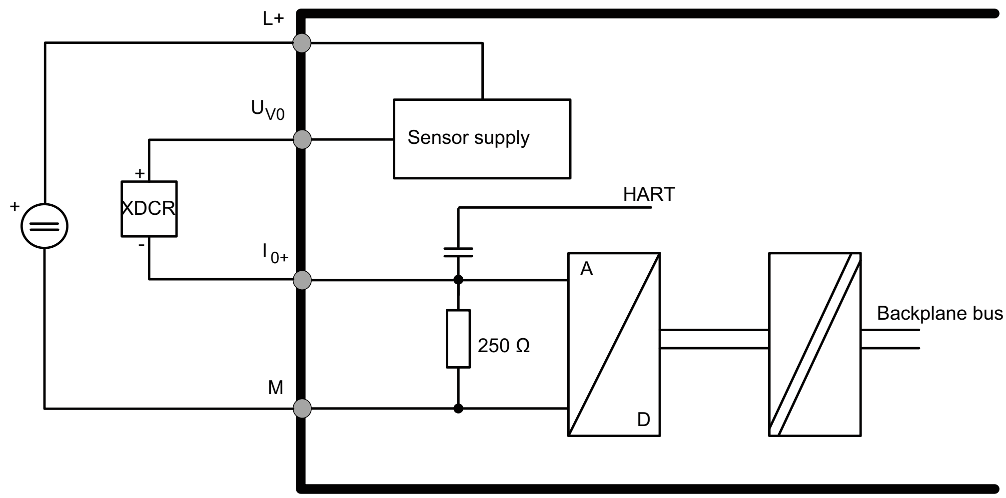
Terminal	Assignment	Terminal	Assignment	Explanations
1	I_0+	2	I_1+	Terminal 1 to 16:
3	I_2+	4	I_3+	I_n+: Input signal "+", channel n
5	I_4+	6	I_5+	Terminal 17 to 32:
7	I_6+	8	I_7+	U_{Vn}: Sensor supply channel n
9	I_{8+}	10	I_{9+}	M: Reference point, ground
11	I_{10+}	12	I_{11+}	
13	I_{12+}	14	I_{13+}	1P1: Supply voltage L+ of the voltage bus 1P
15	I_{14+}	16	I_{15+}	
17	U_{V0}	18	U_{V1}	2P1: Supply voltage L+ of the voltage bus 2P
19	U_{V2}	20	U_{V3}	
21	U_{V4}	22	U_{V5}	1P2: Ground reference of the voltage bus 1P
23	U_{V6}	24	U_{V7}	2P2: Ground reference of the voltage bus 2P
25	U_{V8}	26	U_{V9}	
27	U_{V10}	28	U_{V11}	
29	U_{V12}	30	U_{V13}	
31	U_{V14}	32	U_{V15}	
1P1	L+	1P2	M	
2P1[1]	L+	2P2	M	



[1] If the module is plugged into a TB45R-P32 terminal block suitable for IO redundancy, the potential at this terminal is 1P3.

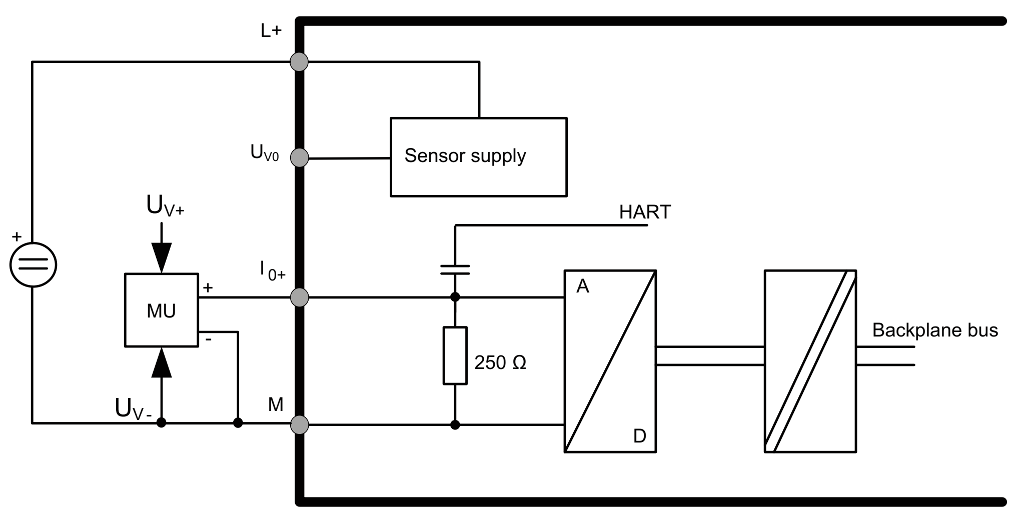
4.3 Connection scheme for transmitter

Example of 2-wire connection of a transmitter using channel 0:



Example of non-isolated connection of a 4-wire transmitter using channel 0:

You can also supply the 4-wire transmitter via the encoder supply terminal U_{V0}. Reference potential of the supply is M. In this case, monitoring of the sensor supply is available for analog input I_0.



4.4 Schematic circuit diagram

The following figure shows the schematic circuit diagram of the I/O module.

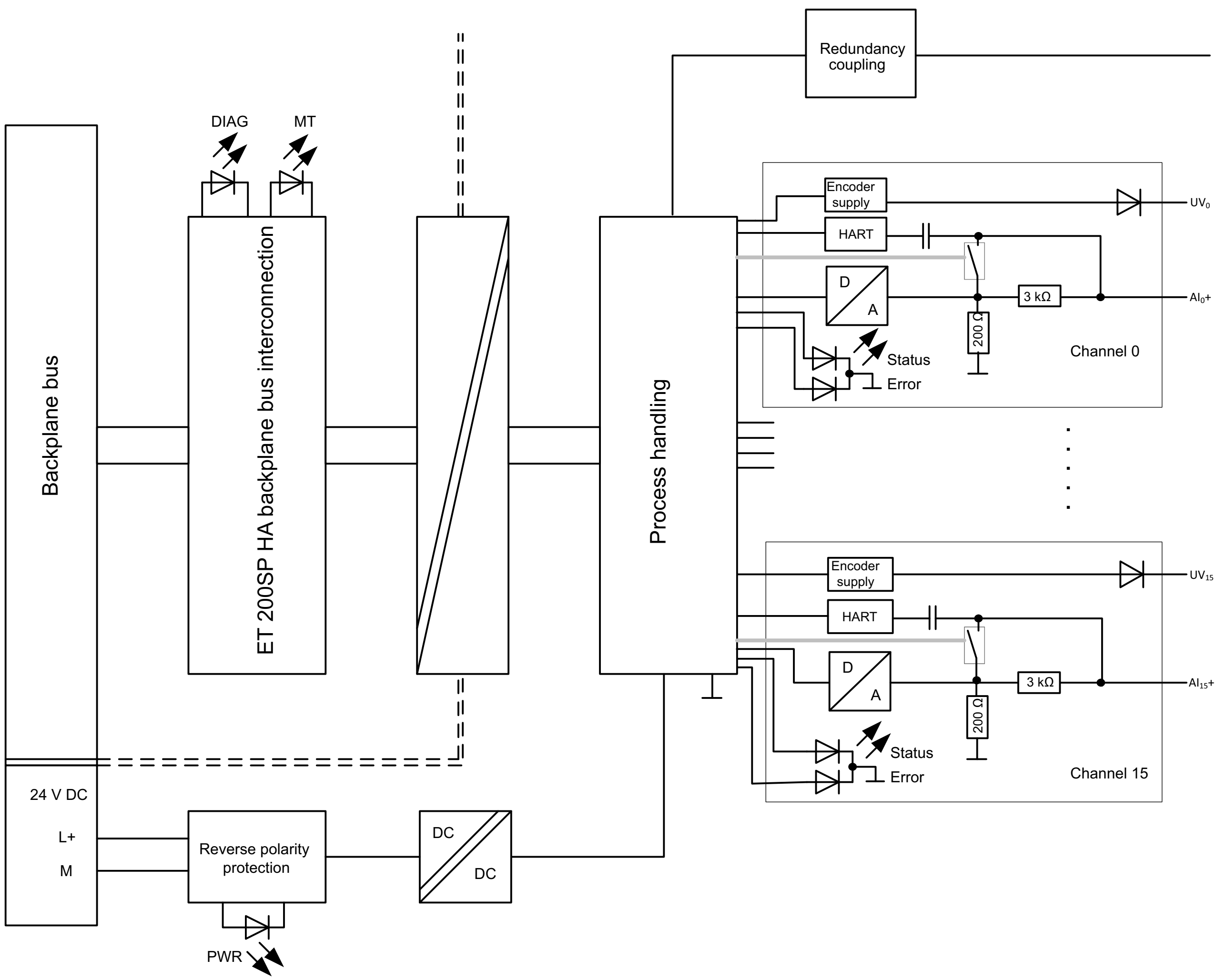


Figure 4-1 Block diagram of AI 16xI 2-wire HART HA

Supply voltage L+/M

Connect the supply voltage L+ to the L+ and M terminals. An internal protective circuit protects the I/O module from reverse polarity. The I/O module monitors whether the supply voltage is connected and present.

Firmware update

The supply voltage L+ must be available on the I/O module at the start of firmware updates and during the update.

Parameters 5

5.1 Configurations of the AI 16xI 2-wire HART HA I/O module

Configuring

You configure the I/O module with PCS 7 V9.0 or higher.

Configuration options

The following configurations, each with value status, are possible:

- Without HART variables in the input area
- With 8 HART variables in the input area
- With a multiHART range in the input/output range

When configuring is done in HW Config, the configuration is carried out indirectly in the parameter assignment dialog of the module.

Note

Only the listed configurations can be directly configured for the I/O module while the CPU is in RUN mode. To set another configuration, first remove the I/O module in CPU RUN. You then add the I/O module with the new configuration while the CPU is in RUN mode. Please note that this might change the I/O address area.

Parameters

You define the functioning of the I/O module via parameters.

The parameters are subdivided into:

- Module/channel parameters (data record 128)
- Parameters that determine the display of HART variables in the address space of the module; HART mapping parameters (data record 130)
- System parameters (potential group, IO redundancy)

See also

Parameter assignment and structure of the module/channel parameters (Page 48)

5.2 Module/channel parameters

Parameters of AI 16xI 2-wire HART HA

You have the following setting options:

Table 5-1 Parameters and their default settings

Parameter	Value range	Default	Parameter reassignment in RUN	Efficiency range
Diagnostics, Missing supply voltage L+	• Enabled • Disabled	Enabled	Yes	Module
IO redundancy	• None • 2 modules	None	Yes	Module
Potential group	• Use potential group of the left module • Allow new potential group	Use potential group of the left module	No	Module
Measurement type	• Disabled • Current (2-wire transducer)	Current (2-wire transducer)	Yes	Channel
Measuring range	• 0 to 10 mA • 0 to 20 mA • 4 to 20 mA • 4 to 20 mA HART There is no underrange in the ranges 0 to 10 mA or 0 to 20 mA That is, the smallest analog value is zero.	4 to 20 mA HART	Yes	Channel
Diagnostics encoder supply	• Enabled • Disabled	Enabled	Yes	Channel
Diagnostic short-circuit to L+	• Enabled • Disabled	Enabled	Yes	Channel
Diagnostics overflow	• Enabled • Disabled	Enabled	Yes	Channel
Diagnostics underflow	• Enabled • Disabled	Enabled	Yes	Channel
Diagnostics, Wire break	• Enabled • Disabled	Enabled	Yes	Channel
Diagnostics HART	• Enabled • Disabled	Enabled	Yes	Channel

Parameter	Value range	Default	Parameter reassignment in RUN	Efficiency range
Smoothing	• None • Weak • Medium • Strong	None	Yes	Channel
Interference frequency suppression	• 60 Hz (integration time 16.6 ms) • 50 Hz (integration time 20 ms) • 10 Hz (integration time 100 ms) When 50 Hz is set, the interference signals of 400 Hz are also filtered automatically.	10 Hz	Yes	Channel
Failure monitoring according to NE43 (As of firmware V1.1)	• Enabled • Disabled	Disabled	Yes	Channel
Wire break limit (firmware V1.0)	• 1.185 mA • 3.6 mA	1.185 mA	Yes	Channel
Number of HART frame repetitions	0 to 10	5	Yes	Channel
Hardware interrupt high limit 1/2	• Enabled • Disabled	Disabled	Yes	Channel
Hardware interrupt low limit 1/2	• Enabled • Disabled	Disabled	Yes	Channel
High limit 1/2	Value	20 mA	Yes	Channel
Low limit 1/2	Value	4 mA at 4 to 20 mA 1 uA for 0 to 10 mA and 0 to 20 mA	Yes	Channel

Note

Unused channels

"Disable" unused channels in the parameter assignment to improve the cycle time of the module.

A disabled channel always supplies the analog value $7FFF_H$.

See also

Parameter assignment and structure of the module/channel parameters (Page 48)

5.3 Explanation of the module/channel parameters

Diagnostics, Missing supply voltage L+

Enabling of the diagnostics for missing or insufficient supply voltage L+.

IO redundancy

You can configure two identical modules redundantly. To do so, plug both modules into a redundant terminal block side by side. You can find additional information on mounting modules in an IO redundancy configuration in the ET 200SP HA System Manual, section "Installing", "Installing terminal block".

In IO redundancy mode, the left module is the master and the right module is the slave.

Both redundant modules measure simultaneously and independently in the process. Both redundant modules generate diagnostics, alarms and process values. The process values are based on the measured values of the respective leading channel that measures with low resistance in the process. By default, the channels of the master are the leading channels. When a module fails or when the failure of a channel is detected, the respective leading channels are switched over to the partner module. Once an error has been corrected, there is no changeover of the leading channels involved.

HART communication can only take place over the leading channel.

Measurement type/measuring ranges

The I/O module supports the following measuring ranges:

Measurement type	Measuring range	Resolution for interference frequency suppression		
		10 Hz	50 Hz	60 Hz
Disabled	--	--	--	--
Current (2-wire transducer)	0 to 10 mA	16 bits including sign	16 bits including sign	15 bits including sign
	0 to 20 mA	16 bits including sign	16 bits including sign	16 bits including sign
	4 to 20 mA			
	4 to 20 mA HART			

The resolution of the analog values is dependent on the interference frequency suppression setting.

HART mode is only possible with a measuring range of 4 to 20 mA. HART communication is not interrupted even at currents below 4 mA.

An overview of the measuring range and the overflow, overrange, etc. can be found in section Analog value display (Page 77).

Diagnostics encoder supply

The sensor supply of the analog input is monitored for overload and short-circuit to M.

Diagnostic short-circuit to L+

Enabling of the monitoring for a short-circuit of the analog input to sensor supply or L+.

The short-circuit and overflow diagnostics can be activated at the same time. If both diagnostics occur simultaneously, the short-circuit diagnostics is output.

Diagnostics, Wire break

You can enable a wire break diagnostics for measuring range 4 to 20 mA or 4 to 20 mA HART. The diagnostics is triggered when the current flow falls below the value for the specific measuring range.

Note the dependency on the configurable wire break limit (for V1.0) or on enabling failure monitoring according to NE43 (as of V1.1).

If you enable wire break and underflow diagnostics simultaneously, the wire break diagnostic is output.

Diagnostics overflow

Enables diagnostics when the measured value reaches the overflow, i.e. exceeds the overrange or nominal range.

Diagnostics underflow

You can enable a underflow diagnostics for measuring range 4 to 20 mA or 4 to 20 mA HART. Diagnostics is triggered when the measured value reaches the underflow, i.e. falls below the underrange or nominal range.

There is no underflow in the ranges 0 to 20 mA or 0 to 10 mA.

Diagnostics HART

Enabling of the diagnostics of the HART frame-specific monitoring and the status information supplied by the connected field device in the HART frame (HART device status).

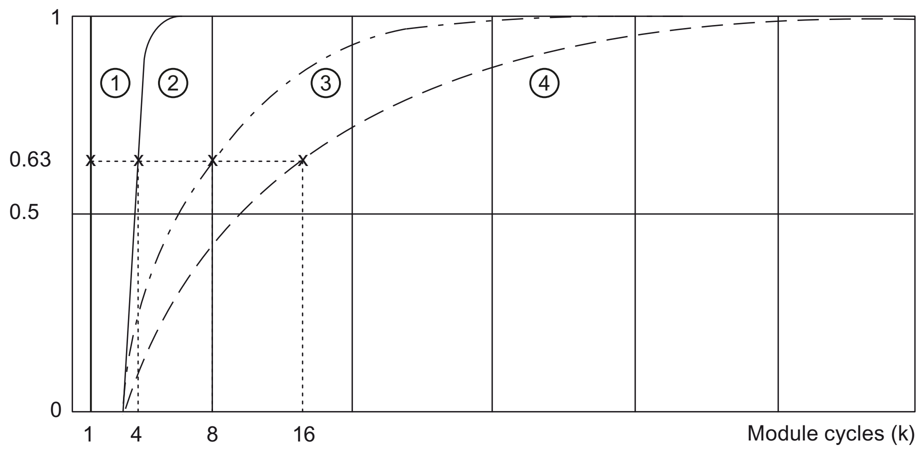
The HART diagnostics are signaled as maintenance events.

Smoothing

The individual measured values are smoothed by filtering. Smoothing can be set in 4 levels.

Smoothing time = number of module cycles (k) x cycle time of the module.

The following figure shows the number of module cycles after which the smoothed analog value approaches 100%, depending of the smoothing setting. This applies to every signal change at the analog input.



① No smoothing (k = 1)

② Weak (k = 4)

③ Medium (k = 8)

④ Strong (k = 16)

Figure 5-1 Smoothing of the analog value

Interference frequency suppression

Suppresses the interference that is caused by the frequency of the alternating voltage network used.

When HART operation is activated, an interference frequency suppression of 10 Hz is recommended so that the HART signal does not influence the analog value.

Failure monitoring according to NE43 (as of V1.1)

The failure monitoring according to NE43 can only be enabled for the measuring range 4...20 mA (4...20 mA HART).

Enabling the failure monitoring changes the settings for monitoring the analog value.

- Disabled
  The limits according to the S7 specifications apply to the monitoring of the analog value
- Enabled
  The limits for monitoring the analog value according to NE43

Information on the measuring ranges and the display with disabled or enabled failure monitoring can be found in the section "Analog value display (Page 77)".

- If failure monitoring is enabled, the following applies to the measuring range:
  3.6 mA < valid measuring range (nominal range) < 21 mA
  - Hysteresis at the upper end of the measuring range (detection "Overflow/nominal range"):
    - The analog value is invalid as of 21 mA.
    - The analog value becomes valid again when it falls below the value 20.5 mA.
  - Hysteresis at the lower end of the measuring range (detection "Underflow/nominal range"):
    - The analog value is invalid as of 3.6 mA.
    - The analog value becomes valid again when it rises above the value 3.8 mA.
- Diagnostics for enabling failure monitoring
  You can enable a diagnostic message for the overflow/underflow (Diagnostics overflow/ Diagnostics underflow).
  If you enable wire break and underflow diagnostics simultaneously, the wire break diagnostic is output.

Note

Wire break limit in V1.0 compatible operation

The I/O module supports compatible operation as of V1.1. If the I/O module is configured as a V1.0 variant with firmware V1.1 or higher, it supports the configurable wire break limit of V1.0 instead of failure monitoring.

Wire break limit (with V1.0)

You use this to specify the current limit for detecting wire breaks. When the current is below the configured wire break limit, the measured value is declared invalid and, when wire break diagnostics is enabled, a corresponding diagnostics is generated. The value status of the analog input is set to "bad".

- With a wire break limit of 3.6 mA, the measured value becomes invalid when the current falls below 3.6 mA and is marked as valid again when it rises above 3.8 mA.
- There is no hysteresis for a wire break limit of 1.185 mA.

Number of HART frame repetitions

Specifies the number of HART frame repetitions. If AI 16xI 2-wire HART HA does not receive a response or receives a faulty response to a HART frame sent to the field device, the frame is repeated, i.e. resent to the field device, according to this setting.

Hardware interrupt 1 / 2

Enabling of a hardware interrupt when the high limit 1 /2 is exceeded or the low limit 1 / 2 is fallen below.

Low limit 1 / 2

Specify a threshold which triggers a hardware interrupt when violated.

High limit 1/2

Specify a threshold which triggers a hardware interrupt when violated.

Potential group

A potential group consists of a group of adjacently placed I/O modules within an ET 200SP HA station that are supplied using a common supply voltage.

A potential group begins with a light terminal block via which the needed supply voltage for all I/O modules of the potential group is fed. The light-colored terminal block interrupts the self-assembling voltage buses to the left neighbor

All other I/O modules of this potential group are plugged into dark terminal blocks. I/O modules on dark terminal blocks take the potentials of self-assembling voltage rails from the left neighbor.

A potential group ends with the dark terminal block that is followed by a light terminal block or a server module in the station configuration.

You can find additional information on the configuration of the potential group in the system manual *SIMATIC; Distributed I/O System; ET 200SP HA*.

5.4 HART mapping parameters

A maximum of 8 HART variables can be configured (mapped) to the address space of the I/O module with the HART mapping parameters.

If you want to use a maximum of 8 HART variables in the input area of the I/O module, configure these parameters in the properties dialog of the I/O module.

Parameters

Table 5-2 HART mapping parameters of the I/O module

<table>
<tr><th colspan="2">Parameter</th><th>Value range</th><th>Default</th><th>Parameter reassignment in RUN</th></tr>
<tr><td rowspan="2">Variable 0</td><td>Channel</td><td>0…15</td><td>0</td><td>Yes</td></tr>
<tr><td>Type</td><td>• Non / CiR
• Primary
• Secondary
• Tertiary
• Quaternary</td><td>Non / CiR</td><td>Yes</td></tr>
<tr><td colspan="5">:
:</td></tr>
<tr><td rowspan="2">Variable 7</td><td>Channel</td><td>0…15</td><td>0</td><td>Yes</td></tr>
<tr><td>Type</td><td>• Non / CiR
• Primary
• Secondary
• Tertiary
• Quaternary</td><td>Non / CiR</td><td>Yes</td></tr>
</table>

See also

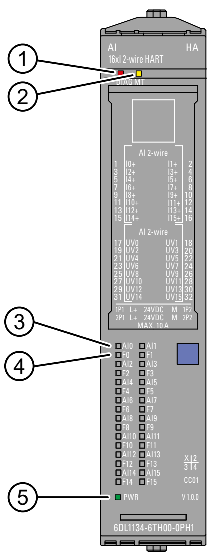
Parameter assignment and structure of the HART mapping parameters (Page 52)

Displays and interrupts 6

6.1 Status and error displays of the AI 16xI 2-wire HART HA I/O module

LED displays

The figure below shows the status and error displays of I/O module.



① DIAG LED (green/red)

② MT LED (yellow)

③ Channel status LED (green)

④ Channel fault LED (red)

⑤ PWR LED (green)

Figure 6-1 LED displays

Meaning of the LED displays

The following tables show the meaning of the status and fault displays. Corrective measures for diagnostics interrupts can be found in the section Diagnostics messages and maintenance events (Page 53).

DIAG LED

Table 6-1 Diagnostics display of the DIAG LED

DIAG LED	Meaning
Off	The supply voltage of the ET 200SP HA is disrupted or switched off.
Flashes	Module is not configured.
On	Module is configured, a diagnostics message is not pending.
Flashes	Module is configured, at least one diagnostic message is pending.

MT LED

Table 6-2 Maintenance display of the MT LED

MT LED	Meaning
Off	No maintenance required.
On	At least one maintenance requirement, i.e. at least one maintenance event has occurred.

Channel status/channel error LED

The green channel status LED indicates whether a channel is enabled. The LED is off when channel or module diagnostics for an enabled channel are pending.

The red channel fault LED lights up when a channel diagnostics are pending with enabled channel.

Table 6-3 Status and error displays of the channel status/channel fault LEDs

Channel status LED	Channel fault LED	Meaning
Off	Off	• Channel disabled or module switched off • Module diagnostics pending
On	Off	Channel enabled and no channel/module diagnostics pending
Off	On	Channel enabled and channel diagnostics pending

PWR LED

Table 6-4 Status display of the PWR LED

PWR LED	Meaning
□ Off	Supply voltage L+ missing
■ On	Supply voltage L+ present

6.2 Interrupts

The I/O module supports diagnostics interrupts and hardware interrupts. You can find additional information about hardware interrupts in appendix "Hardware interrupts (Page 57)".

Diagnostics interrupts

Diagnostics interrupts are used by the I/O module to signal diagnostics messages as well as maintenance events, see also appendix Diagnostics messages and maintenance events (Page 53).

The I/O module generates a diagnostics interrupt at the following events:

- Overtemperature
- Channel/component temporarily unavailable
- Short-circuit / overload of sensor supply
- Wire break
- Low limit violated (underflow)
- High limit violated (overflow)
- Supply voltage missing
- Hardware interrupt lost
- Module is faulty
- Short-circuit of an analog input to sensor supply or L+
- Retentive memory in carrier module defective
- Retentive memory in the terminal block defective
- Redundancy partner has different hardware/firmware
- IO redundancy warning
- Redundancy parameter assignment inconsistent
- HART communication error
- HART field device error

Note

The diagnostics "Module is defective" and "Excess temperature" are not cleared again by the module without a restart.

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7 HART function

Definition

"HART" stands for "Highway Addressable Remote Transducer".

Using the HART functionality, you can exchange data between the I/O module and the connected field devices. The HART protocol is generally accepted as a standard protocol for communication with intelligent field devices: HART is a registered trademark of the HART Communication Foundation (HCF), which owns all the rights to the HART protocol. You can find detailed information about HART in the HART specification.

Advantages of HART

Use of the AI 16xI 2-wire HART HA gives you the following advantages:

- Connection compatibility with standard analog modules: Current loop 4 - 20 mA
- Additional digital communication via the HART protocol, HART communication (Rev. 5 to Rev. 7)
- Numerous field devices with HART functions are in use
- I/O module enables application of HART devices on an IO device based on ET 200SP HA

Use in the system

The I/O module is used in an IO device based on the ET 200SP HA.

You can connect a HART field device to each channel (monodrop mode).

The I/O module serves as HART master, the field devices as HART slaves.

The maximum 16 channels communicate in parallel.

7.1 How HART works

Introduction

The HART protocol describes the physical form of the transfer: transfer procedures, message structure, data formats and commands.

HART signal

The figure below shows the analog signal with the modulated HART signal (FSK method), which consists of sine waves of 1200 Hz and 2200 Hz and has a mean value of 0. It can be filtered out using an input filter so that the original analog signal is available again.

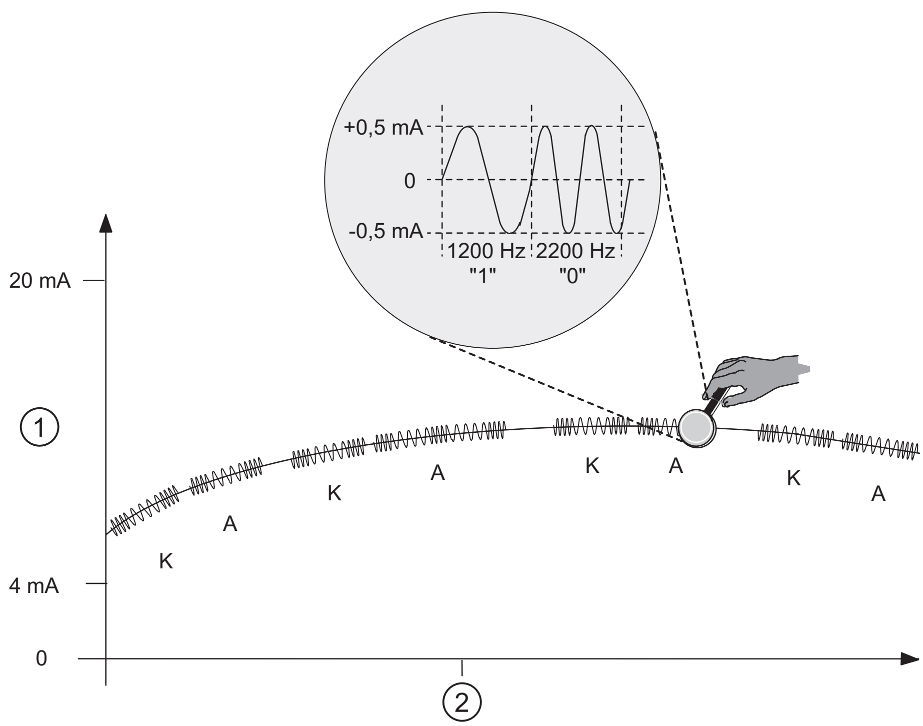


Figure 7-1 The HART signal

①	Analog signal
②	Time (seconds)
K	Command
A	Response

HART communication

The I/O module processes the HART communication in parallel operation, that is, all channels simultaneously.

With enabled HART operation, the I/O module automatically sends HART commands to the connected field devices. These commands always alternate with any pending external HART commands for the specific channel that arrive via the command interface of the module, see section HART command interface (Page 63).

Commissioning a HART field device

Only HART field devices that have been set to short frame address 0 can be operated. If a HART field device with a different short frame address is connected or a connected field device is reconfigured to a short frame address other than 0 during operation, the module starts a scan of all possible short frame addresses at the next re-establishment of HART communication (command 0 with short frame addresses 1...63). As soon as the connected field device responds, it is converted to the short frame address 0 (HART command 6) by the module. The I/O module signals a HART communication error during the scan.

HART commands

The configurable properties of the HART field devices (HART parameters) can be set with HART commands and read out using HART responses. The HART commands and their parameters are divided into three groups with the following properties:

- Universal
- Common practice
- Device-specific

Universal commands must be supported by all manufacturers of HART field devices and common practice commands should be supported. There are also device-specific commands that apply only to the particular field device.

Examples of HART commands

The following two tables show examples of HART commands:

Table 7-1 Examples of universal commands

Command	Function
0	Reads manufacturer and device type - only with this command 0 can field devices be addressed by means of a short frame address
11	Reads manufacturer and device type
1	Reads primary variable and unit
2	Reads current and percentage of range, digitally as floating-point number (IEEE 754)
3	Reads up to four pre-defined dynamic variables (primary variables, secondary variables, etc.)
13, 18	Reads or writes process tag name ("tag"), description and date (dates are also sent)

Table 7-2 Examples of common practice commands

Command	Function
36	Sets high range limit
37	Sets low range limit
41	Perform self-test
43	Sets the primary variable to zero

Structure of the HART protocol

Each HART frame sent from the I/O module to the connected field device (request frame) and each HART frame received by the field device (response frame) has the following basic structure.

PREAMBLE	STRT	ADDR	COM	BCNT	STATUS	DATA	CHK

PREAMBLE: Bytes (0xFF) for synchronizing,
default: 5 bytes (can be changed using DS131 - DS138)

STRT: Start character (start delimiter)

ADDR: Address of the field device (1 byte; short address or 5 bytes; long address)

COM: HART command number

BCNT: Byte count, number of bytes to follow without checksum

STATUS: HART device status (1st and 2nd status byte). Only present for a response frame. For structure of HART device status, see below.

DATA: Transferred user data / parameters, quantity depending on command (0…230 bytes)

CHK: Checksum

With the exception of the preamble bytes, this structure is contained in the communication data of the HART command interface. See section HART job and response data records (Page 65).

HART responses always contain data. Status information (HART device status; 1st and 2nd status bytes) is always sent together with a HART response. status bytes) is always sent together with a HART response. You should evaluate these to make sure the response is correct.

Structure of HART device status (1st and 2nd status bytes)

Table 7-3 1st status byte

Bit 7 = 1: "Communication error"	
Bit 6 = 1 Bit 5 = 1 Bit 4 = 1 Bit 3 = 1 Bit 2 = 0 Bit 1 = 1 Bit 0 = 0	Parity error Overflow Framing error Checksum error Reserved Overflow in the receive buffer Reserved
Bit 7 = 0: "No communication error" Bit 0...6: "Specific according to the response frame"	

Table 7-4 2nd status byte

Bit 7 = 1	Device fault
Bit 6 = 1	Configuration changed
Bit 5 = 1	Startup (cold start)
Bit 4 = 1	Additional status information available
Bit 3 = 1	Fixed analog output current setting
Bit 2 = 1	Analog output current saturated
Bit 1 = 1	Secondary variable outside the limits
Bit 0 = 1	Primary variable outside the range

HART fast mode

The I/O module supports the processing of HART commands as an SHC string ("Successive HART Command").
If a HART command with set SHC bit is detected by the I/O module for a channel, this channel is reserved for external HART commands for approximately 2 s. Internal HART commands are not issued, see section HART command interface (Page 63)

Burst mode

The I/O module does not support burst mode. HART commands with set burst bit are ignored and are not forwarded to the connected field device.

7.2 HART applications

System environment for the use of HART

To use an intelligent field device with HART functionality, you require the following system environment:

- 4 to 20 mA current loop
- Connecting measuring transmitters to the I/O module

HART parameter assignment tool "Client":
You can assign the HART parameters using an external handheld operating device (HART Handheld) or a HART configuration tool (PDM). Both assume the function of a "client":

The parameter assignment tool affects the entire I/O module; the HART handheld is connected in parallel to the field device.

PDM (Process Device Manager) is available standalone or integrated in STEP 7 HW Config. In the latter case, PDM is opened from HW Config.

HART system integration:
The I/O module takes on the function of a "master" by receiving commands from the HART parameter assignment tool, for example, forwarding them to the smart field device and sending back the responses. The interface of the I/O module is made up of data records that are transmitted internally via the ET 200SP HA IO device between IM and I/O module. The data records must be created and interpreted by the client.

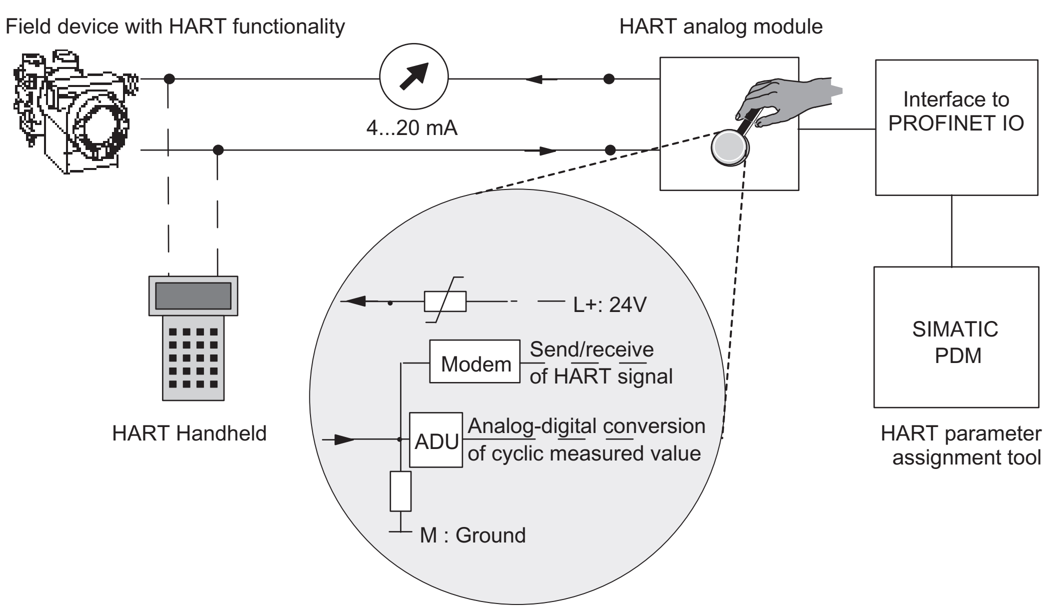


Figure 7-2 System environment for the use of HART

Error management

The two HART status bytes (HART device status) that are transferred with every response of the field device contain error indications regarding the HART communication, HART command and device status.
These are evaluated by the I/O module and made available in the system using S7 maintenance messages.

Configuration/commissioning

How to configure the I/O module with HW Config You assign parameters for the individual channels with regard to the actual analog value acquisition and the use of HART variables in the input address space of the module.
You can configure one field device per channel. The configuration/parameter assignment of the connected field device is then carried out from this configured field device using PDM or the EDD for the ET 200SP HA.

Parameter reassignment of the field devices

The I/O module generally accepts triggered parameter reassignments for field devices. The awarding of access rights can only be made in the parameter assignment tool.

To reassign parameters of the field devices connected to the I/O module, proceed as follows:

1. You initiate parameter reassignment of a field device with a HART command that you enter with the SIMATIC PDM parameter assignment tool.
2. After you complete parameter reassignment of a HART field device, the corresponding bit is set in the HART device status of the connected field device (in the 2nd status byte).
3. As a result of field device reconfiguration, the I/O module triggers a "Configuration changed" maintenance message if such a message is enabled. This maintenance message should be regarded as a notice and not an error. It is automatically deleted again by the I/O module after approximately 1 minute.

If enabled, a maintenance message can also be triggered by a new parameter assignment with the handheld device.

See also

Pin assignment of the AI 16xI 2-wire HART HA I/O module (Page 12)

Diagnostics messages and maintenance events (Page 53)

7.3 HART variables

Introduction

Numerous HART field devices make available additional measured quantities (e.g. sensor temperature).

A maximum of four HART variables supported by the connected field device are read cyclically for each channel with enabled HART functionality. The HART variables are read automatically via the HART command 3 (for field devices with HART Rev. 5 and 6) or via command 9 (for field devices with HART Rev. 7 or later).

These four HART variables per channel are always stored in HART variable data record 121 and can be read at any time.

In addition, you can optionally configure the following:

- You can map a maximum of 8 HART variables to the input address space of the I/O module. The HART variables are assigned to a channel in the properties dialog of the I/O module. This allows you to easily process measured values directly from the field device as input data in the automation device.
- You can configure a multiHART area in the input/output address space of the I/O module. You can read all HART variables available in the I/O module with each of these multiHART areas. You specify the multiHART area in the properties dialog for the I/O module.

Address assignment

Each HART variable occupies 5 bytes of input data. As soon as you configure (map) at least one HART variable in the input address space, the addresses for all 8 tags are assigned (40 bytes).

If the multiHART range is used, an additional 6 bytes of input address space and 1 byte of output address space is allocated.

Configuration of HART variables

You can configure up to 8 HART variables in the properties dialog of the I/O module. You select these from the four HART variables provided by each channel:

- PV (Primary Variable)
- SV (Secondary Variable)
- TV (Tertiary Variable)
- QV (Quaternary Variable)

When HART mode is enabled, the I/O module cyclically reads the variables provided by the connected field devices itself and makes them available in the configured input address space.

Configuring of a multiHART area

You can use a multiHART area for the AI 16xI 2-wire HART HA I/O module. You select whether or not to use a multiHART area in the properties dialog of the I/O module.

When HART mode is enabled, the I/O module cyclically reads the variables provided by the connected field devices itself. You can request and read one of these HART variables via the multiHART range.

Quality code

The quality code describes the process status of the corresponding HART variable.

Basic structure of the quality code

Bit	7 to 6	5....2	1 to 0
	Quality 0 0: Bad 0 1: Uncertain 1 0: Good 1 1: Good	Sub-status Coded according to "PROFIBUS PA Profile for Process Control Devices"	Limits 0 0: OK 0 1: Low limit 1 0: High limit 1 1: Constant

The quality codes generated by the I/O module conform to the HART revision of the field device used.

For field devices with HART Revision 5 and 6

The quality code is formed exclusively from the 1st and 2nd status byte (HART device status) of the response frames (HART command 3).

Quality code	Meaning (process status)	
80_H	Value is okay	Also applies when the following bits are set in the 2nd status byte of the HART response frame: • Configuration changed • Startup (cold start) • Fixed analog output current setting
78_H	Value is uncertain	Also applies when the following bits are set in the 2nd status byte of the HART response frame: • Additional status information available • Analog output current saturated • Secondary variable outside the limits • Primary variable outside the range
84_H	Response code RC8: Update error	
24_H	Response code RC16: Access restricted	Request from field device refused
23_H	Communication error or HART variable not present in the field device	

Quality code	Meaning (process status)	
37_H	Initialization value from analog module	After module start-up, after incorrect operation of the multiHART interface or after redundancy failover to the standby channel[1]
00_H	Initialization value from S7 system	

[1] A channel is in standby mode when a redundancy failover occurred due to an error in an IO-redundant module.

For field devices as of HART revision 7

The quality code is formed from the 1st status byte (HART device status) and the "Device variable status" (DVS) of the response frames (HART command 9).

Quality code	Meaning (process status)	
80_H	Value is okay	
89_H	"Good" with "low limit"	Process status, formed from the "Device variable status" (DVS) of the response frames with corresponding limits (see above).
$8A_H$	"Good" with "high limit"	
28_H...$2B_H$	"Bad"	
68_H...$6B_H$	"Poor accuracy"	
78_H...$7B_H$	"Manual" or "Fixed" (manually controlled or fixed value)	
88_H...$8B_H$	"More device variable state available" (additional status information available)	
84_H	Response code RC8: Update error	
24_H	Response code RC16: Access restricted	Request from field device refused
23_H	Communication error or HART variable not present in the field device	
37_H	Initialization value from analog module	After module start-up, after incorrect operation of the multiHART interface or after redundancy failover to the standby channel[1]
40_H	Read alternatively via command 3	
00_H	Initialization value from S7 system	

[1] A channel is in standby mode when a redundancy failover occurred due to an error in an IO-redundant module.

See also

HART variable data record (Page 73)

Parameter assignment and structure of the HART mapping parameters (Page 52)

8 Technical specifications

Technical data of AI 16xI 2-wire HART HA

Note

Power supply

The supply and input voltages of the ET200SP HA system must always be generated with voltage/current supplies with safe electrical isolation (SELV/PELV according to IEC/UL61010-2-201) with a rated value of 24 V DC ±20 %.

Article number	6DL1134-6TH00-0PH1
General information	
Product type designation	AI 16 x I 2-wire mA HART
Firmware version	V1.1
• FW update possible	Yes
Usable terminal block	TB type H1, M1 and N0
Color code for module-specific color identification plate	CC01
Product function	
• I&M data	Yes; I&M0 to I&M3
Engineering with	
• PCS 7 configurable/integrated as of version	V9.0
Redundancy	
• Redundancy capability	Yes; With TB type M1
CiR – Configuration in RUN	
Reparameterization possible in RUN	Yes
Supply voltage	
Rated value (DC)	24 V
permissible range, lower limit (DC)	19.2 V
permissible range, upper limit (DC)	28.8 V
Reverse polarity protection	Yes
Input current	
Current consumption (rated value)	80 mA; without sensor supply
Current consumption, max.	90 mA; without sensor supply
24 V encoder supply	
• 24 V	Yes
• Short-circuit protection	Yes; Electronic (response threshold 0.7 A to 1.5 A)
• Output current per channel, max.	0.5 A
• Output current per module, max.	2 A
Power loss	
Power loss, typ.	4.5 W; without sensor supply
Address area	
Address space per module	
• Address space per module, max.	34 byte; 32-byte inputs and 2 bytes for QI information
• Address space per module with HART, max.	74 byte; 32-byte inputs and 2 bytes for QI information, 40-byte inputs for HART
• Address space per module with MultiHART, max.	41 byte; 32-byte inputs for HART and 2 bytes for QI information, 6-byte inputs for HART, and 1-byte output for MultiHART command
Analog inputs	
Number of analog inputs	16
permissible input current for current input (destruction limit), max.	30 mA
Input ranges (rated values), currents	

Article number	6DL1134-6TH00-0PH1
• 0 to 20 mA	Yes; 16 bit incl. sign
• Input resistance (0 to 20 mA)	250 Ω
• 4 mA to 20 mA	Yes; 16 bit incl. sign
• Input resistance (4 mA to 20 mA)	250 Ω
Cable length	
• shielded, max.	800 m; With unshielded cables up to 800 m, remember that (external) EMC loads can cause incorrect measured values.
Analog value generation for the inputs	
Measurement principle	integrating (Sigma-Delta)
Integration and conversion time/resolution per channel	
• Resolution with overrange (bit including sign), max.	16 bit; 15 bit at 0 ... 10 mA and 60 Hz interference suppression
• Integration time, parameterizable	Yes; channel by channel
Smoothing of measured values	
• parameterizable	Yes; none, weak, medium, strong, channel-by-channel
Encoder	
Connection of signal encoders	
• for current measurement as 2-wire transducer	Yes
Errors/accuracies	
Linearity error (relative to input range), (+/-)	0.01 %
Temperature error (relative to input range), (+/-)	0.005 %/K
Crosstalk between the inputs, min.	60 dB
Repeat accuracy in steady state at 25 °C (relative to input range), (+/-)	0.05 %
Operational error limit in overall temperature range	
• Current, relative to input range, (+/-)	0.5 %
Basic error limit (operational limit at 25 °C)	
• Current, relative to input range, (+/-)	0.1 %
Interrupts/diagnostics/status information	
Diagnostics function	Yes
Alarms	
• Diagnostic alarm	Yes
• Limit value alarm	Yes; two upper and two lower limit values in each case
Diagnostic messages	
• Monitoring the supply voltage	Yes
• Wire-break	Yes; channel by channel
• Short-circuit	Yes; Channel-by-channel, short-circuit of the encoder supply to ground or of an input to the encoder supply
• Overflow/underflow	Yes; channel by channel

Article number	6DL1134-6TH00-0PH1
Diagnostics indication LED	
• MAINT LED	Yes; Yellow LED
• Monitoring of the supply voltage (PWR-LED)	Yes; Green PWR LED
• Channel status display	Yes; Green LED
• for channel diagnostics	Yes; Red LED
• for module diagnostics	Yes; green/red DIAG LED
Potential separation	
Potential separation channels	
• between the channels	No
• between the channels and backplane bus	Yes
• Between the channels and load voltage L+	No
Isolation	
Isolation tested with	1 500 V DC/1 min, type test
Ambient conditions	
Ambient temperature during operation	
• horizontal installation, min.	-40 °C
• horizontal installation, max.	70 °C; Observe derating
• vertical installation, min.	-40 °C
• vertical installation, max.	60 °C; Observe derating
Dimensions	
Width	22.5 mm
Height	115 mm
Depth	138 mm
Weights	
Weight, approx.	148 g

Note

With use of sensor supply U_{Vn} (terminal 17...32), the following derating must be observed:

Horizontal mounting position:

Up to 70°C, up to 12 channels when using the encoder supplies (total current of outputs 300 mA)

Up to 70°C, up to 16 channels when using the encoder supplies

Up to 65°C, up to 16 channels when using the encoder supplies (total current of outputs 400 mA)

Up to 60°C, up to 16 channels when using the encoder supplies (total current of outputs 2 A)

Vertical mounting position:

Up to 60°C, up to 12 channels when using the encoder supplies (total current of outputs 300 mA)

Up to 60°C, up to 16 channels without using the encoder supplies

Up to 55°C, up to 16 channels when using the encoder supplies (total current of outputs 400 mA)

Up to 50°C, up to 16 channels when using the encoder supplies (total current of outputs 2 A)

Maximum supply voltage ripple during HART operation

- 390 mVpp at 50 Hz
- 8.3 mVpp at 300 Hz

Cycle time

The cycle time describes the time slice in which signals from the inputs are acquired and processed.

The cycle time depends on the parameter assignment and can be different for each channel.

Operating mode	Cycle time
Analog inputs for 60 Hz interference frequency suppression	16.6 ms
Analog inputs for 50 Hz interference frequency suppression	20 ms
Analog inputs for 10 Hz interference frequency suppression	100 ms

Redundancy failover time

At an IO redundancy failover, the update time of process values is delayed once by a time less than or equal to this time.
The maximum redundancy failover time of the I/O module is 20 ms + cycle time.

Drivers, parameters, diagnostics messages and address space

A

A.1 Concept of the driver and diagnostics blocks

Use with PCS 7

The following information shows you the benefits of using PCS 7. PCS 7 automatically carries out many important parameter assignments and interconnections of blocks.

If you want to use the ET 200SP HA outside of the PCS 7 environment, pay attention to the next sections.

Tasks of the driver and diagnostics blocks (driver blocks)

In process control systems, certain demands are placed on the diagnostics/signal processing. This includes the monitoring of modules, DP/PA slaves and DP master systems for malfunctions and failures.

To enable this, blocks are available in the PCS 7 library that implement the interface to the hardware including test functions.

These blocks perform two basic tasks:

- They provide signals from the process to the AS for further processing.
- They monitor modules, DP/PA slaves, and DP master systems for failure.

When the process signals are read in, these blocks access the process image input (or process image partition) (PII) and when the process signals are output, they access the process image output (or process image partition) (PIQ).

Concept

The concept of the driver and diagnostics blocks for PCS 7 can be characterized as follows:

- The separation between user data processing (CHANNEL blocks) and diagnostics data processing (MODULE blocks)
- The symbolic addressing of the I/O signals
- The automatic generation of the MODULE blocks by CFC

This block concept supports all modules from the list of approved modules.
When new Siemens or non-Siemens module types are integrated, the meta-knowledge for the driver generator can be extended by additional XML files (object and action lists).

Note

Note the following:

- The library with the driver blocks has to installed using the Setup program on the PC. This is the only method of ensuring that the meta-knowledge required for the driver generator is available. You must not copy the library from another computer.
- You can also use driver blocks from another library (for example, your own blocks from your own library). You can specify this additional library in the "Generate module drivers" dialog box. The driver generator then searches for the block to be imported in the library specified here. If the block is not found here, it is searched for in the library specified in the control file (XML file).
- If the S7 program contains a signal-processing block but not from one of the PCS 7 libraries, you have to specify the version of the driver library from which the driver blocks are to be imported in the "Generate module drivers" dialog box.

Time-optimized processing

To enable time-optimized processing during runtime, the organization blocks for error handling (for example, OB85, OB86) are automatically divided into runtime groups and the driver blocks are integrated in the corresponding runtime groups.

If an error occurs, the SUBNET block, for example, activates the relevant runtime group, the RACK block or MODULE block contained in the runtime group detects the error, evaluates it and outputs a process control message to the OS.

The diagnostics information of the module block (OMODE_xx output) is also transferred to the corresponding CHANNEL block (MODE input). If necessary, this information can be displayed by means of a PCS 7 block that can be operated and monitored on the OS or by means of a user block in a process picture (color change of the measured value or flashing display, etc.).

A.2 Parameters, diagnostic messages and address space without PCS 7

A.2.1 Parameter assignment

Parameter assignment in the user program

You can reconfigure individual channels of the I/O module and HART variable mapping in RUN without this affecting the other channels.

Changing parameters in RUN

The parameters are transferred to the I/O module with the instruction "WRREC".

- Module/channel parameters using data record 128
- HART mapping using data record 130

The parameters assigned with STEP 7 are not changed permanently in the CPU, which means the parameters assigned with STEP 7 are valid again after a restart.

Output parameter STATUS

If errors occur when transferring parameters with the "WRREC" instruction, the module continues operation with the previous parameter assignment. The STATUS output parameter contains a corresponding error code.

The STATUS output parameter is 4 bytes long and is configured as followed:

- Byte 1: Function_Num, general error code
- Byte 2: Error_Decode, location of the error code
- Byte 3: Error_Code_1, error code
- Byte 4: Error_Code_2, manufacturer-specific extension of the error code

Module-specific errors are displayed via Error_Decode = 0x80 and Error_Code_1 / Error_Code_2.

Error_Code_1	Error_Code_2	Cause	Remedy
0xB0	0x00	Number of the data record unknown	Enter valid number for data record.
0xB1	0x01	Length of the data record is incorrect	Enter valid value for data record length.
0xB2	Variable	Module cannot be reached	• Check station - is the module plugged in correctly? • Check the parameters of the WRREC instruction.
0xE0	0x01	Incorrect version in the header	Correct the version number of the parameter block.
0xE0	0x02	Header error (number or length of parameter block)	Correct the length and number of the parameter blocks.

Error_Code_1	Error_Code_2	Cause	Remedy
0xE1	0x01	Reserved bit set	Check and correct the parameters
0xE1	0x02	Invalid diagnostics enable bit set for operating mode	
0xE1	0x04	Invalid value for hardware interrupt limit	
0xE1	0x08	Invalid coding for interference frequency suppression / integration time	
0xE1	0x09	Invalid coding for smoothing	
0xE1	0x0E	Invalid redundancy parameter assignment	
0xE1	0x10	Invalid measurement type	
0xE1	0x11	Invalid measuring range	
0xE1	0x21	Incorrect number of HART repetitions	

Valid parameters

Only the values specified in the following are permitted. Values that are not listed are rejected by the I/O module.

Each parameter data record is checked by the I/O module. If a parameter with errors is detected, the complete data record is rejected and the parameters of the I/O module remain unchanged.

A.2.2 Parameter assignment and structure of the module/channel parameters

Structure of data record 128

Data record 128 has a length of 236 bytes.

It contains module parameters and all channel or technology parameters of the 16 channels.

The channel and technology parameters are divided into parameters that influence the actual analog value acquisition, diagnostics enables and basic parameters of the HART communication.

You can specify and change other parameters and HART-specific settings with data records 131 to 146. See section HART-specific settings (Page 74).

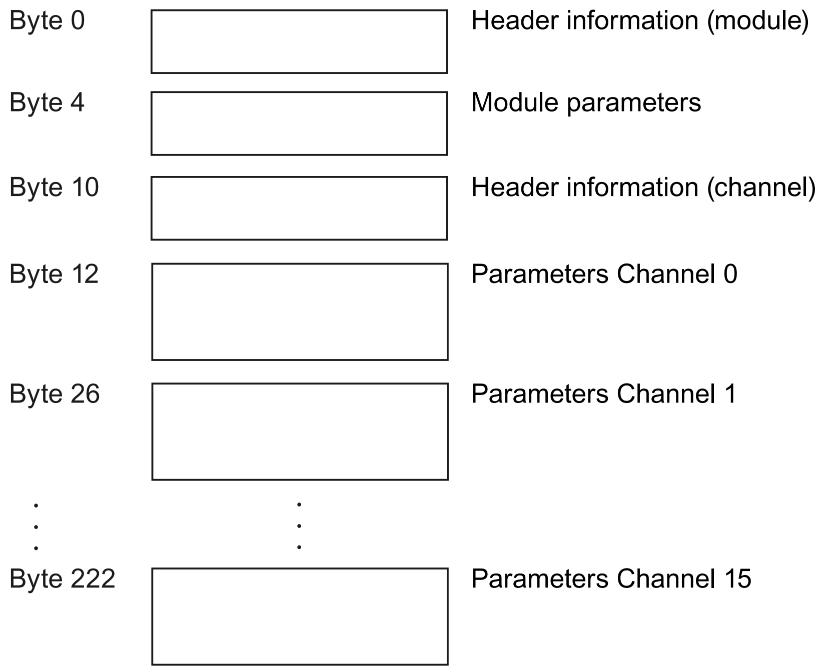


Figure A-1 Structure of data record 128

Header information and module parameters

The following figure shows the structure of the header information and module parameters.

	7	6	5	4	3	2	1	0	
Byte 0	0	1	0	0	0	0	0	0	Bits 3–0: Minor version Bits 5–4: Major version
Byte 1	0	0	0	0	0	0	1	0	Number of subsequent parameter structures
Byte 2	0	0	0	0	0	0	0	1	Number of subsequent module parameter blocks
Byte 3	0	0	0	0	0	1	1	0	Length of the module parameter block
Byte 4	0	0			0	0	0		Bit 0: Diagnostics for missing supply voltage L+ Bits 5–4: Redundancy role: 0="off" 2="Master (station-internal)" 3="Slave (station-internal)"
Byte 5	0	0	0	0	0	0	0	0	Reserved
Byte 6									Standalone mode: irrelevant Redundancy mode: Logical base address, high byte
Byte 7									Standalone mode: irrelevant Redundancy mode: Logical base address, low byte
Byte 8									Standalone mode: irrelevant Redundancy mode: Logical base address of the partner module, high byte
Byte 9									Standalone mode: irrelevant Redundancy mode: Logical base address of the partner module, low byte
Byte 10	0	0	0	1	0	0	0	0	Number of subsequent channel parameter blocks = 16
Byte 11	0	0	0	0	1	1	1	0	Length of a channel parameter block = 14

Figure A-2 Header information and module parameters of data record 128

Channel parameters

Each channel parameter block contains the channel parameters for the analog output including HART (14 bytes).

The following figure shows the structure of the channel parameters for channel 0 to 15.

x = 12 + (channel number * 14); with channel number 0…15

All unused bits and the bits or bytes marked as "reserved" must be set to zero.
You activate a channel parameter by setting the corresponding bit to "1" or the corresponding value.

Byte x (bits 7 6 5 4 3 2 1 0)

Measurement type
0 = Analog input disabled
3 = Current (2-wire transducer)

Byte x+1 (bits 7 6 5 4 3 2 1 0)

Measuring range
2 = 0...20 mA
3 = 4...20 mA
7 = 4...20 mA HART
8 = 0...10 mA

Byte x+2 (bits 7 6 5 4 3 2 1 0)

Bits 0 to 3: Smoothing:
0000 = None
0001 = Weak
0010 = Medium
0011 = Strong

Bits 4 to 7: Interference frequency suppression:
0001 = 60 Hz
0010 = 50 Hz
0011 = 10 Hz

Byte x+3 (bits 7 6 5 4 3 2 1 0; bit 3 = 0, bit 0 = 0)

Bit 1: Enable diagnostics HART

Bit 2: Failure monitoring according to NE43 (as of firmware version 1.1.0)
1 = Enabled
0 = Disabled
Wire break limit (up to firmware version lower than 1.1.0)
0 = 1.185 mA
1 = 3.6 mA with hysteresis

Bits 4 to 7: Hardware interrupts
Bit 4 = 1 when high limit 2 is violated
Bit 5 = 1 when low limit 2 is violated
Bit 6 = 1 when high limit 1 is violated
Bit 7 = 1 when low limit 1 is violated

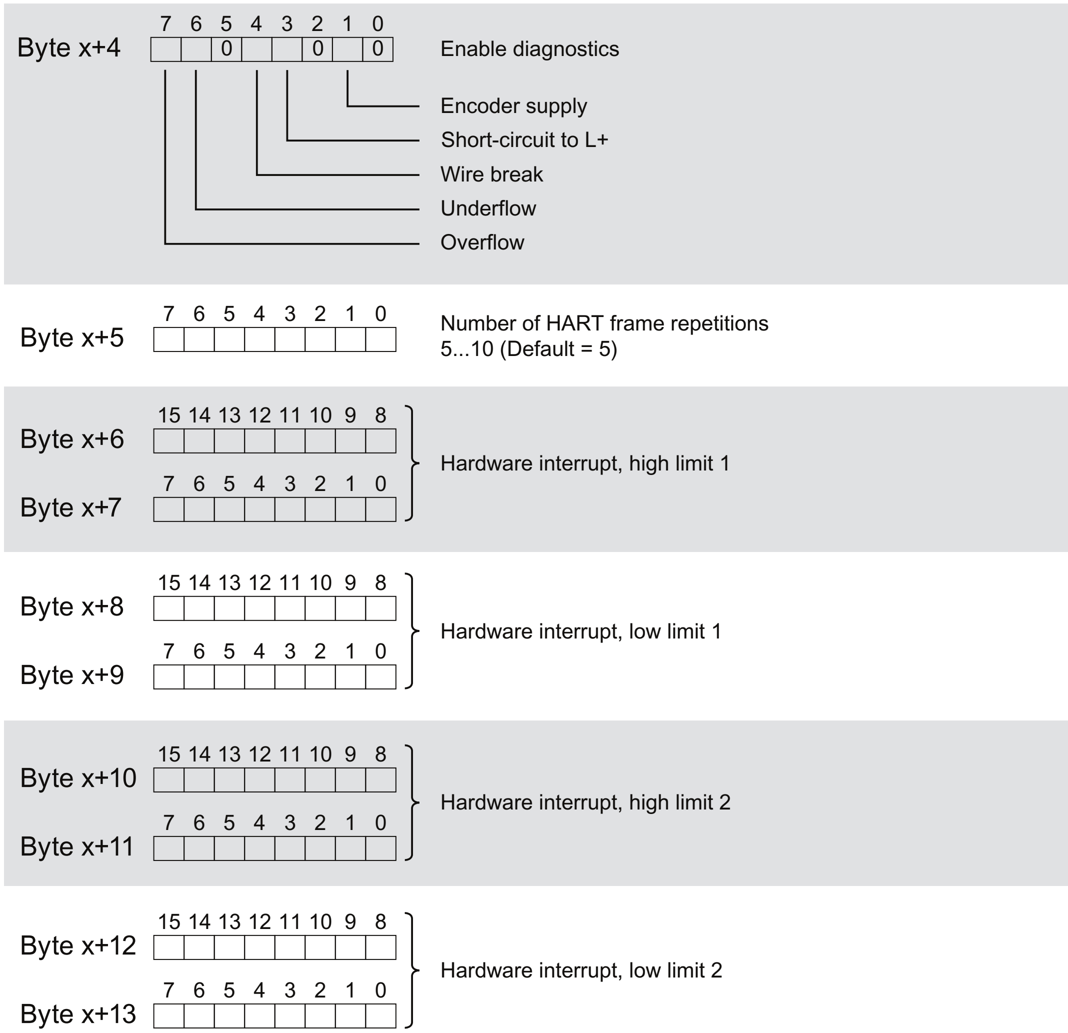


Figure A-3 Structure of byte x to x+n for channels 0 to 15

A.2.3 Parameter assignment and structure of the HART mapping parameters

Structure of data record 130

Data record 130 has a total length of 20 bytes.

You can use the parameters of data record 130 to configure/map up to 8 HART variables of the individual channels in the input address space of the I/O module, provided you have selected configuration with HART variables in the input area, see section Configurations of the AI 16xI 2-wire HART HA I/O module (Page 15).

Header information

The figure below shows the structure of the header information.

	Bits	
Byte 0	7 6 5 4 3 2 1 0	Version = 0
Byte 1	7 6 5 4 3 2 1 0	Block length = 19
Byte 2	7 6 5 4 3 2 1 0	Block type = 75
Byte 3	7 6 5 4 3 2 1 0	Reserved = 0

Figure A-4 Header information of data record 130

Parameters

The figure below shows the parameter assignment of the 8 HART variables.

x = 4, 6, 8, 10, …, 18

	Bits	
Byte x	7 6 5 4 3 2 1 0	Channel number: 0...15
Byte x+1	7 6 5 4 3 2 1 0	Variable identifier: 1 = Primary 2 = Secondary 3 = Tertiary 4 = Quaternary 15 = non / CiR

Figure A-5 Parameters of data record 130

The variable ID 15 = non/CiR prevents the I/O module from configuring/mappping a HART variable; that is, the corresponding memory area in the input address area remains unallocated.

A.2.4 Diagnostics messages and maintenance events

Diagnostics messages

A diagnostics message is generated for each detected diagnostics event. The DIAG LED on the module flashes.

There is additionally a channel-specific display of the diagnostics through the corresponding channel fault/channel status LEDs.

The diagnostics messages can, for example, be read out in the diagnostics buffer of the CPU.

Diagnostic messages are assigned either to one input on a channel-specific basis or to all inputs as a module message. In the event of diagnostics messages that affect the entire module, all channels are switched off. For diagnostics messages that relate to individual channels, only the corresponding analog input is affected.

Table A-1 Diagnostic messages, their meaning and possible remedies

Diagnostics message	**Error code**	**Assignment**	**Meaning**	**Remedy**
Channel/component temporarily unavailable	$1F_H$	Module	Update of the firmware is being performed or has been canceled. The module does not perform any measurements during this time.	• Restart firmware update • Wait for firmware update
Overtemperature	5_H	Module	• Ambient temperature is too high • Short-circuit or overload of one or more digital outputs/sensor supplies	• Check wiring • Eliminate the cause of the excess temperature and restart the module
Wire break	6_H	Analog input	• Impedance of encoder circuitry is too high • Wire break between the module and sensor • Channel not connected (open)	• Use a different encoder type or modify the wiring, for example, using cables with a larger cross-section • Connect the cable • Disable diagnostics • Connect the encoder contacts
High limit violated	7_H	Analog input	• The analog value is above the overrange. • Short-circuit of input to L+	Correct the module/encoder tuning
Low limit violated	8_H	Analog input	The analog value is below the underrange.	Correct the module/encoder tuning
Parameter assignment error	10_H	Module	• Parameter assignment does not match the terminal block used • Incorrect parameter assignment	• Correct the parameter assignment • Check terminal block[1]
Supply voltage missing	11_H	Module	Missing or insufficient supply voltage L+	• Check wiring of the supply voltage L+ on the terminal block • Check the terminal block type
Hardware interrupt lost	16_H	Module	Error is reported because the events occur faster than they can be processed in the system	
Module is faulty	100_H	Module	Internal module error has occurred	Replace module
Short-circuit to L+	105_H	Analog input	Short-circuit of the analog input to sensor supply or L+	Correct the process wiring

Diagnostics message	Error code	Assignment	Meaning	Remedy
Short-circuit/overload at external encoder supply	$10E_H$	Analog input	• Short-circuit of the encoder supply to ground • Encoder supply overload	• Correct the module/ encoder tuning • Check the wiring
Inconsistent parameter assignment	123_H	Module	Error in the redundancy parameters transferred to the module	• Check the parameter assignment • Check the slot and terminal block
Retentive memory in carrier module defective	154_H	Module	A memory block error was detected on the support module	Replace carrier module
Retentive memory in the terminal block defective	155_H	Module	A memory block error was detected on the terminal block	Replace terminal block

[1] Only the terminal blocks (TB 45R ...) are suitable for use with I/O modules in IO redundancy.

Maintenance events

A maintenance event is generated each time a maintenance requirement is identified. The MT LED lights up on the module.

Maintenance messages are assigned either to one input on a channel-specific basis (HART error) or to inputs as a module message affecting all inputs.

Maintenance messages have no direct effect on the function of the module or the analog value acquisition. Maintenance events of the HART communication do not affect the analog value acquisition of the module.

Table A-2 Maintenance messages, their meaning and possible remedies

Maintenance message	Error code	Assignment	Meaning / Cause	Remedy
Redundancy partner has different hardware/firmware version	120_H	Module	The redundantly configured and interconnected I/O modules are not compatible	Check and replace modules
IO redundancy warning	121_H	Module	Unable to correctly communicate with the partner module	• Check/replace right module • Check/replace left module • Check/replace terminal block See also system manual *Distributed I/O System; ET 200SP HA*", section "Replacing an I/O module for I/O redundancy"

Maintenance message	Error code	Assignment	Meaning / Cause	Remedy
HART communication error	141_H	Analog input	• HART field device is not responding • Timing error • HART field device did not understand the command that was sent (1st status byte)	• Check the process wiring • Correct the parameter assignment • Set current ≥4 mA • Increase number of configured repetitions
HART primary variable outside of limits	142_H	Analog input	• Incorrect parameters in the HART field device • HART field device is set to "Primary variable outside of limits" in simulation mode • Incorrect measuring point • Parameter assignment of primary variable outside of limits	• Check the parameter assignment of the HART device • Correct the simulation • Check whether the correct measuring transducer is connected
HART analog output current of the field device is saturated	143_H	Analog input	• The output current of the HART field device is saturated: • Incorrect parameters in the HART field device • HART field device is set to a measured value that is too high in simulation mode • Incorrect measuring point	
HART output current of the field device specified	144_H	Analog input	• The output current of the HART field device is fixed: • Incorrect parameters in the HART field device • HART field device is set to a measured value that is too high in simulation mode • Incorrect measuring point	
HART additional status information available	145_H	Analog input	In the HART device status (in the 2nd status byte), the HART field device identifier for "further status information available" has been set	Read out status with HART command 48 and clear error/cause if necessary
HART configuration changed[1]	146_H	Analog input	In the HART device status (in the 2nd status byte), the identifier for "parameter reassignment" of the HART field device has been set	If no diagnostics interrupt is to be triggered by reconfiguration, the HART diagnostics must be disabled.
HART malfunction in the field device	147_H	Analog input	In the HART device status (in the 2nd status byte) the field device reports a malfunction	• Read out status with HART command 48 and clear error/cause if necessary • Replace the field device

Maintenance message	Error code	Assignment	Meaning / Cause	Remedy
HART secondary variable outside the limits	149_H	Analog input	• Incorrect parameters in the HART field device • HART field device is set to "Non-primary variable outside limits" in simulation mode • Incorrect measuring point • Parameter assignment of non-primary variable is outside the limits	• Check the parameter assignment of the HART device • Correct the simulation • Check whether the correct measuring transducer is connected
Retentive memory in carrier module defective	154_H	Module	Fault detected in the data block on the carrier module during operation	Replace carrier module
Retentive memory in the terminal block defective	155_H	Module	A memory block error was detected on the terminal block during operation	Replace terminal block

[1] **Response of the HART configuration changed maintenance message**

If the HART field device signals "parameter reassignment" (configuration changed) in the 2nd status byte, the module generates the maintenance message "HART configuration changed". If the field device withdraws the message in the 2nd status byte within a minute, the maintenance message is also deleted again by the module. If the message in the 2nd status byte is still set after a minute, the module then independently sends HART command 38 for resetting the message in the field device.

A.2.5 Hardware interrupts

Hardware interrupts

The I/O module can initiate hardware interrupts when limit violations of analog input signals occur.

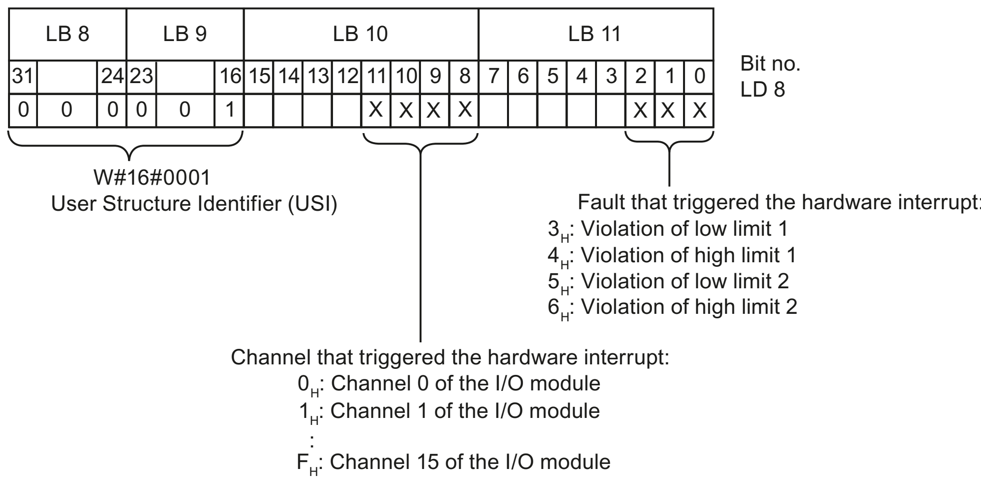
Interrupt OBs are called automatically if an interrupt occurs.

You can find detailed information on the event in the hardware interrupt organization block with the "RALRM" instruction (read additional interrupt information).

The module generates a hardware interrupt at the following events:

- Violation of the low limit 1
- Violation of the high limit 1
- Violation of the low limit 2
- Violation of the high limit 2

The module analog input (channel) that has triggered the hardware interrupt is entered in variable OB40_POINT_ADDR in the start information of OB40. The figure below shows the assignment to the bits of local data double word 8.



A.2.6 Address space

Abbreviations

- "IB" stands for input byte, that is, the module start address in the input area
- "QB" stands for output byte, that is the module start address in the output area
- "QAIn" stands for value status (QI) of analog input n
- "QC"stands for Quality Code

Evaluating the value status

The input address space contains one value status (QI) bit for each analog input.

Irrespective of the diagnostics enables, each value status (each QI bit) provides information on the validity of the corresponding process value.

- Value status = 1: Process value okay, "good"
- Value status = 0: Process value not okay, "bad"

Basically, the value status is set to "good" when the analog value can be acquired without error. The value status is set to "bad" in the following cases:

- The analog value cannot be acquired due to an error.
- The analog input is disabled.

Address space

The following tables show the allocation of the address space of the I/O module.

Addresses for the HART area are only available if HART variables or a multiHART area were configured in the properties dialog of the module.

Input area

IB x +	7	6	5	4	3	2	1	0
0…1	Analog value, analog input 0							
: :	: :							
30…31	Analog value analog input 15							
32	QAI7	QAI6	QAI5	QAI4	QAI3	QAI2	QAI1	QAI0
33	QAI15	QAI14	QAI13	QAI12	QAI11	QAI10	QAI9	QAI8
34	HART range							
: :								

HART range

When HART variables or the multiHART area is used, the HART area is arranged directly flush against the QI bits of the analog inputs.
The structure of the HART area is dependent on the configuration:

- In the configuration with 8 HART variables in the input area, the area has a length of 40 bytes and always contains 8 HART variables, each with a 4-byte value and a 1-byte quality code (QC).

IB x +	HART range	
34 : 37	Value	Configured HART variable 0
38	QC	
39 : 42	Value	Configured HART variable 1
43	QC	
: : :		
69 : 72	Value	Configured HART variable 7
73	QC	

- In the configuration with 1 multiHART area in the input/output area, the area has a length of 6 bytes:

IB x +	HART range		
34	Acknowledgment		multiHART range
35 : 38	Value	HART variable	
39	QC		

Output range

In the configuration with a multiHART area, there is one byte in the output area.

QB x	Command	multiHART range

Evaluating HART variables

If you have configured (mapped) HART variables for the I/O module, then 8 HART variables with 5 bytes each are stored in the input address space.

Each HART variable consists of a 4-byte real value and a 1-byte quality code. The quality code describes the validity of the value, see section HART variables (Page 36).

The assigned HART variables are automatically updated by the module and can be used directly in the user program.

Evaluating the multiHART area

You can select the multiHART area in the parameter assignment dialog.

The multiHART range occupies 1 byte in the output range (command) and 6 bytes in the input range (1 byte acknowledgment + 5 bytes HART variable).

You can read all HART variables available in the module using the multiHART area. To do so, you must use the command byte in the output area to request a HART variable from the module.

The command byte identifies the requested HART variable (HART variable reference):

HART variable reference	
Bit 0…3: Variable identifier 1 = Primary 2 = Secondary 3 = Tertiary 4 = Quaternary	Bit 4...7: Channel number 0...15

The command is acknowledged via the acknowledgment byte in the input range. As soon as the HART variable reference requested using the command can be read, the requested HART variable can be evaluated.

If you leave the command byte of the multiHART range unchanged, the module will continuously update the corresponding HART variable.

If an invalid HART variable reference is requested, this is also acknowledged accordingly. The value of the corresponding HART variable is then zero and the quality code is signaled with $0x37_H$ (initialization value of the analog module).

Example:
The primary variable is to be read from channel 6.

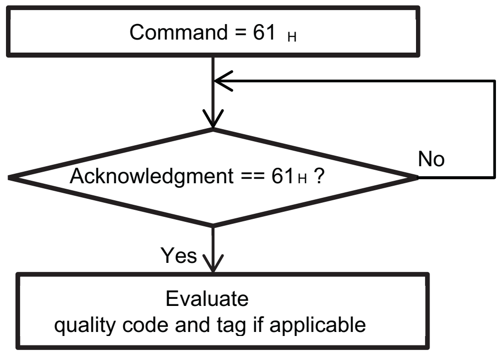


Figure A-6 Command = 0x61

HART operating data records B

B.1 HART command interface

Data records

The HART commands are sent as so-called external HART requests from the client (e.g. PDM) to the connected field device using data records. The response of the field device is provided again in the system using data records.

The HART communication may only be handled by one client per channel. If a channel is handled by several clients, the response made available by the I/O module cannot be allocated to one client with certainty. The I/O module does not support client management.

Rules

- After having written a request data record, a client must read the response data record before it may write another request data record.
- The client can evaluate the "processing status" in the response data record: If the "processing status" indicates "successful" or "error," the response data record contains current response data or error indications, respectively.
- The response data record must always be read completely because the I/O module may modify the data record after the initial reading with "successful" or "error" status.
  If the processing status in the response data record indicates "successful" or "error", the data record contains current response data or error displays.
- The client may only write a request data record to the I/O module again when it has read the response to the previously written request data record via the corresponding response data record. Otherwise, the response from the I/O module is overwritten.
- The STATUS component in the response frame (HART device status in the response data) provides information on whether errors have occurred and, if so, which errors.

Each request is stored on a channel-specific basis, and the corresponding request data record is locked. Another writing of the same request data record is thus not possible and is acknowledged with BUSY.
The lock of the request data record is reset after the termination or completion of the requested HART command.

SHC sequence

If a HART command with set SHC bit is sent to the I/O module, this channel is reserved for HART commands for 2 seconds. That is, no more internal HART commands are sent to the field device for this channel.

For each additional HART command with SHC bit set, the I/O module reserves this channel again for HART commands for an additional 2 seconds. Command 3 or 9 for reading the HART variables starts being sent cyclically to the field device again if a HART command without a set

SHC bit is detected for this channel, or if no further command is received for the channel within 2 seconds of the previous HART command.

Reading/writing data in RUN

HART operating data records are transferred to the module with the instruction "WRREC" and read by the module with the instruction "RDREC".

Errors during the transfer are indicated at output parameter STATUS of the "WRREC" or "RDREC".

The following HART operating data records are available:

Data record number	Description	Length (bytes)	Writable	Readable
80	HART job channel 0	240	Yes	Yes
81	HART response channel 0	240	No	Yes
82	HART job channel 1	240	Yes	Yes
83	HART response channel 1	240	No	Yes
: :				
110	HART job channel 15	240	Yes	Yes
111	HART response channel 15	240	No	Yes
121	HART variables	320	No	Yes
131	HART parameter channel 0	8	Yes	Yes
132	HART parameter channel 1	8	Yes	Yes
: :				
146	HART parameter channel 15	8	Yes	Yes
148	HART directory	17	No	Yes
149	HART feature data	3	No	Yes

B.2 HART job and response data records

HART commands are handled on a channel-specific basis using a separate command interface with one request data record and one response data record each.

Channel	Data record number	
	Request to the field device	Response from the field device
0	80	81
1	82	83
2	84	85
3	86	87
:		
:		
15	110	111

Structure of job data records 80, 82, 84, 86...110

Byte	Meaning	Comment
0	Request control	
1	Number of preamble bytes	5…20, 255
2…239	Communication data according to HART specification	

Coding "Request control":

Bit 0…1: Reserved = 0

Bit 2: 0 = Parameters are not checked

Bit 3...4: Reserved = 0

Bit 5: 0 = Transparent format *
1 = Compact format

Bit 6: 1 = Enable SHC mode

Bit 7: 0 = HART Request

* HART commands are processed by the I/O module in both transparent message format and compact message format. However, the response data from the I/O module is always made available in transparent message format.

Note

When "Number of Preamble Bytes" = 255, the number of preambles set with the parameters is used. The default setting is five. You can reconfigure the number of preamble bytes using the HART-specific settings (see section HART-specific settings (Page 74)).

Structure of response data records 81, 83, 85, 87...111

In case of response error

Byte	Meaning	Comment
0	Response control	
1	HART group error display	
2	Protocol error	
3…239	Response data according to HART specification	Only present when "Response result" = 6 = "Error, with data"

In case of response error

Byte	Meaning	Comment
0	Response control	
1	HART group error display	
2…239	Response data according to HART specification	Only present when "Response result" = 4 = "Successful, with data"

Coding "Response control":

Bits 0-2: Response result (processing status)

0 = Inactive
1 = Inactive (reserved)
2 = Waiting
3 = Waiting, executing
4 = Successful, with data
5 = Successful, without data
6 = Error, with data
7 = Error, without data

Bit 3: 0 = Burst mode not active;

Bit 4: 0 = Response data come directly from the HART device

Bit 5: 0 = Response data in transparent message format

Bit 6: 0 = SHC mode not active
1 = SHC mode active

Bit 7: 0 = HART response

Coding "HART group error display"

Bit number	Meaning	Explanation
0	Additional status information available	(2nd HART status byte) You obtain additional status information, if required, with HART command 48.
1	HART communication error	The field device has detected a communication error when receiving the command. The error information is in the first HART status byte
2	Parameter check	0: HMD parameters unchanged 1: Check HMD parameters

Bit number	Meaning	Explanation
3	Reserved	Always 0
4…7	HART protocol error during response	0: Unspecified error 1: HMD error 2: Channel fault 3: Command error 4: Query error 5: Response error 6: Query rejected 7: Profile query rejected 8: Manufacturer-specific query rejected 9 - 15: Not used

Coding "HART protocol error during response"

HART protocol error during response	Meaning	Explanation
0	Unspecified error	Always 0
1	HMD error	0: Not specified 1: Internal communication error 2: Parameter assignment error 3: HW fault 4: Wait time expired 5: HART timer expired 6…127: Reserved 128...255: Manufacturer-specific
2	Channel fault	0: Not specified 1: Line fault 2: Short-circuit 3: Open line 4: Low current output 5: Parameter assignment error 6…127: Reserved 128…255: Manufacturer-specific
3	Command error	0-127: HART protocol, Bit 7 = Always 0
4	Query error	Bit 0 = 0: Reserved Bit 1 = 1: Receive buffer overflow Bit 2 = 0: Reserved Bit 3 = 1: Checksum error Bit 4 = 1: Framing error Bit 5 = 1: Overflow error Bit 6 = 1: Parity error Bit 7 = 1: Reserved

HART protocol error during response	Meaning	Explanation
5	Response error	Bit 0 = 1: GAP timeout Bit 1 = 1: Receive buffer overflow Bit 2 = 1: Timeout Bit 3 = 1: Checksum error Bit 4 = 1: Framing error Bit 5 = 1: Overflow error Bit 6 = 1: Parity error Bit 7 = 1: Reserved
6	Query rejected	0: Unspecified 1: Compact format not supported 2: SHC not supported 3: Impermissible command 4: No resources 5: Channel in standby operation [1] 6 to 127: Reserved 128 to 255: Manufacturer-specific
7	Profile query rejected	0: Not specified (not supported)
8	Manufacturer-specific query rejected	0: Not specified (not supported)

[1] An external HART request was refused because the channel is not the active channel of a redundancy pair. The request must be sent to the connected field device via the partner module

Example of HART programming (HART command interface)

For HART channel 0, command 01 is to be sent in transparent message format to the HART field device with address "98 CF 38 84 F0".
A positive edge at input 4.0 of a digital input module leads to the writing of the HART command.

The following assumptions are made:

- The module address of the I/O module is 512 (200_H).
- The data record is stored in DB80: starting from address 0.0, length of 11 bytes.
- In this example, DB80 (request data record for channel 0) consists of 11 bytes.

```
                                   Explanation
    A I 4.0
    FP M 101.0
    = M 104.0
m2: CALL SFB53, DB53
    REQ :=M104.0                   Write request
    ID :=DW#16#200                 Module address
    INDEX :=80                     Data record number 80
    LEN :=11                       Length 11 bytes
```

```
DONE :=M51.7
BUSY :=M51.0
ERROR :=M51.6
STATUS :=MD92                     Block status or error information
RECORD :=P#DB80.DBX0.0 BYTE 11    Source area in DB80
A M 51.0
SPB m2
BE
```

Table B-1 DB80: Transparent message format

Byte	Initial value (hex)	Comment (Hex)
0	00	Req_Control (00 = Transparent message format. 40 = Transparent message format with SHC sequence)
1	05	Number of preamble bytes (05-14)
2	82	Start character (02 = Short Frame with command 0) (82 = Long Frame with other commands)
3	98	Address (with command 0, the address is exactly 1 byte long and has the value 0.)
4	CF	
5	38	
6	84	
7	F0	
8	01	Command (CMD)
9	00	Length in bytes
10	98	Checksum (CHK) (calculated as EXOR addition starting from byte 2 "Start character" up to the last byte of the command) The check-sum must not be sent with the job.)

A HART command can also be sent in compact message format. In this case, the data transmitted via DB 80 is reduced to 4 bytes.

Table B-2 DB80: Compact message format

Byte	Initial value (hex)	Comment (Hex)
0	20	Req_Control (20 = Compact message format 60 = Compact message format with SHC string)
1	05	Number of preamble bytes (5…20, 255)
2	01	Command (CMD)
3	00	Length in bytes

You can learn when the response from the field device was received by cyclically reading data record DS81 for HART channel 0. The response is always supplied in transparent message format.

Table B-3 FC81: Reading of the response to DB81 with SFB 52

```
                                          Explanation
m3: CALL SFB52, DB52
    REQ :=M1                              Read request
    ID :=DW#16#200                        Module address
    INDEX :=81                            Data record number 81
    MLEN :=200                            Target length
    VALID :=M49.7
    BUSY :=M49.1
    ERROR :=M49.6
    STATUS :=MD100                        Block status or error information
    LEN :=MW104
    RECORD :=P#DB81.DBX0.0 BYTE 200       Target area in DB81
    A M 49.1
    SPB m3
    BE
```

The program part A M 49.1 to SPB m3 is only required if reading is to occur within a block cycle.

As long as the processing status (byte 0 of DB81) is at 3 (waiting, executing), the response has not yet been received from the field device. As soon as the processing status changes to greater than 3, the HART request is finished.

With a processing status of 4, the request finished without errors and the response data can be evaluated.

With a processing status of 5, the request also finished without errors but without response data from the field device.

With a processing status of 6 or 7, the request finished with errors. You can find more detailed information in byte 1 of DB81 (see table "HART group fault display") and for a HART protocol error also in byte 2 of DB81 (see table "HART protocol error during response").

B.3 HART directory

Structure of the HART directory

Byte	Meaning	Comment
0	Profile Revision Number	= 2, 0 (Revision 2.0)
1		
2	Index of Client Management	= 255 (not relevant)
3	Number of Clients	= 1
4	Number of Channels	= 16
5	Write Read Index Offset	= 1 (The response to a job data record is made with the data record number of the request data record + 1)
6	Index of HMD Feature Parameter	= 149
7	Index of HMD Module Parameter	= 255 (not relevant)
8	Start Index of Burst Buffer Area	= 255 (not relevant)
9+n	Index of HMD Channel Parameter (Channel n)	= 131+n
9+n+4	Index of HART Client Channel Message Data	= 80+(2*n) The HART request data sets are not configurable. Data sets starting from data set number 80 (80, 82, 84, 86,...) are always used.

B.4 HART feature data

Structure of the HART feature data

Byte	Meaning	Comment
0	Byte 0	= 0x62 Bit 1 = 1: "Parameter check result is given with a read response" Bit 5 = 1: "Compact format is supported" Bit 6 = 1: "SHC mode is supported"
1	Byte 1	= 0
2	Max Length Data Unit	= 240 (maximum length of the HART request data records)

B.5 HART variable data record

The I/O module supports a maximum of 4 HART variables per channel in enabled HART operation. These variables are read cyclically, provided they are supported by the connected field device. These 64 HART variables are available to be read in HART variable data record 121.

Each HART variable consists of a 4-byte real value and one byte of quality code.

In IO redundancy mode, the HART variables are only updated if the HART interface is on the corresponding module/channel. The channel may not be in standby mode. If the HART interface is on the partner module/partner channel, the corresponding HART variable is initialized (quality code = 0x37).

Structure of the HART variable data record

Byte	Meaning	
Channel 0		
0…3	Value	Primary Variable (PV)
4	Quality code	
5…8	Value	Secondary Variable (SV)
9	Quality code	
10…13	Value	Tertiary Variable (TV)
14	Quality code	
15…18	Value	Quaternary Variable (QV)
19	Quality code	
Channel 1		
20…39	HART variables same as for channel 0	
:		
:		
Channel 15		
300...319	HART variables same as for channel 0	

If HART is not enabled or the respective HART variable is not supplied from the connected field device, the corresponding variable = 0 and the QC = 0x37 (initialization value from the analog module).

B.6 HART-specific settings

HART communication is available via standard parameter assignment.

You can specify additional HART-specific settings on a channel-specific basis using data records 131…146.

The parameters assigned with STEP 7 are not changed permanently in the CPU, which means the parameters assigned with STEP 7 are valid again after a restart.

Every new parameter assignment of the analog module resets the HART-specific settings back to the initial values from parameter data record 128.

The I/O module does not apply the HART-specific settings if there is missing supply voltage L +.

If you make changes in IO redundancy mode using the HART-specific settings, these changes only apply to the addressed module or the addressed channel. There is no synchronization with the partner channel.

Channel	Data record number
0	131
1	132
2	133
3	134
:	
:	
15	146

Structure of the HART-specific settings

Byte	Bits 7 6 5 4 3 2 1 0	Description
Byte 0	Must be 128	Bit 7 is reset after evaluation of data record by module
Byte 1	Must be 5	Offset for manufacturer-specific parameters according to HART specification
Byte 2		Number of HART repetitions (0...10) Initial value from parameter data record 128
Byte 3		Number of HART preamble bytes (0, 5...20, 255) Initial value = 5*
Byte 4	Must be 0	Field device mode according to HART specification
Byte 5		Client timeout in s (1...255 s) Initial value = 60 s
Byte 6	0 0 0 0 0 0 (bits 7 to 2) Bit 1: 1 = Primary master, 0 = Secondary master Bit 0: 1 = Enable HART, 0 = Disable HART	Initial value corresponding to HART activation from parameter data record 128 Always as primary master
Byte 7	Must be 0	Reserved

Figure B-1 Settings

* When the number of HART preamble bytes = 0, the number of preamble bytes required by the connected field device are used, but no fewer than 5.
When the number of HART preamble bytes = 255, then 20 preamble bytes are used.

Analog value display C

C.1 Analog value display in the current measurement range 0 to 10 mA

Table C-1 Representation / current measuring range 0 to 10 mA

Values		Current measuring range	Range
Dec.	**Hex.**	**0 to 10 mA**	
32767	7FFF	>11.76 mA	Overflow
32511	7EFF	11.76 mA	Overrange
27649	6C01		
27648	6C00	10 mA	Nominal range
20736	5100	7.5 mA	
1	1	361.69 nA	
0	0	0 mA	

See also

Explanation of the module/channel parameters (Page 18)

C.2 Analog value display in the current measurement range 0 to 20 mA

Table C-2 Representation / current measuring ranges 0 to 20 mA

Values		Current measuring range	Range
Dec.	Hex.	0 to 20 mA	
32767	7FFF	>23.52 mA	Overflow
32511	7EFF	23.52 mA	Overrange
27649	6C01		
27648	6C00	20 mA	Nominal range
20736	5100	15 mA	
1	1	723.4 nA	
0	0	0 mA	

C.3 Analog value display with failure monitoring in the current measurement range 4...20 mA (4...20 mA HART)

Table C-3 Representation / current measuring range with failure monitoring according to NE43 (current measuring range 4 to 20 mA and 4 to 20 mA HART)

Values		**NE43**			
hex	mA	Ranges		QI	Interrupt
		Overflow range (upper failure range)	7FFF	bad	Overflow
7EFF	22.81				
72C0	21				
		Hysteresis [1]			
6F60	20.5				
		Measuring range	Process values	good	
6C00	20				
0	4				
FFFF	3.9994				
FEA6	3.8	Hysteresis [2]			
FD4D	3.6	Underflow range (lower failure range)	8000	bad	Underflow / Wire break
ED00	1.185				

[1] Hysteresis at the upper end of the measuring range (detection "Overflow/nominal range"):

- The analog value is invalid as of 21 mA.
- The analog value becomes valid again when it falls below the value 20.5 mA.

[2] Hysteresis at the lower end of the measuring range (detection "Underflow/nominal range"):

- The analog value is invalid as of 3.6 mA.
- The analog value becomes valid again when it rises above the value 3.8 mA.

C.4 Analog value display without failure monitoring in the current measurement range 4...20 mA (4...20 mA HART)

Table C-4 Representation / current measuring range without enabled failure monitoring (S7 definition - current measuring range 4 to 20 mA and 4 to 20 mA HART)

Values		Current measuring range	Range
Dec.	Hex.	4 to 20 mA	
32767	7FFF	>22.81 mA	Overflow
32511	7EFF	22.81 mA	Overrange
27649	6C01		
27648	6C00	20 mA	Nominal range
20736	5100	16 mA	
1	1	4 mA + 578.7 nA	
0	0	4 mA	
-1	FFFF		Underrange
-4864	ED00	1.185 mA	
-32768	8000	< 1.185 mA	Underflow / Wire break

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